Extra Practice 1: Patterns, Sequences and the General Term

A. Skills

1. Classify each sequence as arithmetic, geometric or neither. State d or r when there is one.

(a) $3, 10, 17, 24, \ldots$ (b) $6, 24, 96, 384, \ldots$ (c) $50, 44, 38, 32, \ldots$

(d) $1, 8, 27, 64, \ldots$ (e) $-1, 5, -25, 125, \ldots$ (f) $90, 30, 10, \frac{10}{3}, \ldots$

(g) $2.4, 2.9, 3.4, 3.9, \ldots$ (h) $3, 5, 9, 17, 33, \ldots$ (i) $\frac{1}{4}, \frac{1}{2}, 1, 2, \ldots$

2. Write the next three terms of each sequence.

(a) $-7, -2, 3, 8, \ldots$ (b) $1000, 100, 10, \ldots$ (c) $5, -10, 20, \ldots$

(d) $2, 3, 5, 8, 13, \ldots$ (e) $12, 9.5, 7, \ldots$ (f) $\frac{3}{4}, \frac{3}{2}, 3, \ldots$

3. Write the first five terms of the sequence with the given general term.

(a) $t_n = 9n - 4$ (b) $t_n = 15 - 2n$ (c) $t_n = 4(3)^{n-1}$

(d) $t_n = 2n^2 - 1$ (e) $t_n = 100(0.5)^{n-1}$ (f) $t_n = \dfrac{n}{n+1}$

4. State whether each sequence is finite or infinite. (a) $2, 4, 6, \ldots, 98$ (b) $1, -1, 1, -1, \ldots$ (c) $5, 10, 20, 40, 80$

B. Applying

5. A sequence has general term $t_n = 7n + 12$.

(a) Determine t_{40}.

(b) Which term is equal to 173?

(c) Is 200 a term? Explain.

6. A sequence has general term $t_n = 3(2)^{n-1}$. Which term is equal to 384?

7. A ball is dropped from 81 cm. Each bounce reaches $\frac{2}{3}$ of the height of the bounce before.
(a) Write the heights of the first four bounces.

(b) Is the sequence arithmetic or geometric? Is it a linear or non-linear function?

8. A gym membership costs \$60 to join plus \$35 per month. The total paid after n months forms a sequence.
(a) Write the first four terms.

(b) Write a general term t_n.

(c) After how many months has a member paid \$480 in total?

9. The points $(1, 4)$, $(2, 7)$, $(3, 10)$ and $(4, 13)$ are the first four terms of a sequence. Without graphing, decide whether the points lie on a line, and state the sequence's t_{10}.

10. Which general term gives the sequence $8, 5, 2, -1, \ldots$?
A. $t_n = 3n + 5$ **B.** $t_n = 11 - 3n$ **C.** $t_n = 8 - 3n$ **D.** $t_n = 8(-3)^{n-1}$

11. Numerical Response The first term of a sequence is 2 and each term after that is the previous term times 2 plus 1. The value of t_6 is _____.

C. Stretch

12. The terms $x + 1$, $2x + 3$ and $4x + 1$ are three consecutive terms of an arithmetic sequence. Determine x and the three terms.

13. A sequence starts $t_1 = 1$, $t_2 = 3$, and each term after that is the sum of the two terms before it. Write the first eight terms. Is the sequence arithmetic, geometric or neither?

Extra Practice 2: Arithmetic Sequences

A. Skills

1. Each sequence is arithmetic. State d and the next two terms.
(a) $17, 26, 35, \ldots$ (b) $8, -3, -14, \ldots$ (c) $-2.4, -1.8, -1.2, \ldots$

(d) $\frac{3}{5}, 1, \frac{7}{5}, \ldots$ (e) $2a + 7,\ 5a + 4,\ 8a + 1, \ldots$

2. An arithmetic sequence has $t_4 = -6$ and $d = 2.5$. Write the first five terms.

3. Determine the general term in simplest form and t_{50}.
(a) $6, 11, 16, \ldots$ (b) $90, 82, 74, \ldots$ (c) $-7, -4.5, -2, \ldots$

4. Determine the number of terms in each sequence.
(a) $12, 19, 26, \ldots, 278$ (b) $60, 56.5, 53, \ldots, -10$

5. (a) An arithmetic sequence has $t_1 = 14$ and $d = -3$. Which term is -100?

(b) An arithmetic sequence has $d = 7$ and $t_{30} = 200$. Determine t_1.

(c) An arithmetic sequence has $t_1 = -20$ and $t_{17} = 28$. Determine d.

B. Applying

6. Insert five arithmetic means between -9 and 15.

7. How many multiples of 6 are there between 250 and 740?

8. Toothpicks make a row of triangles: 1 triangle uses 3 toothpicks, 2 triangles use 5, and 3 triangles use 7.
(a) Determine the general term.

(b) How many toothpicks make a row of 60 triangles?

(c) With 250 toothpicks, what is the longest row of triangles that can be made?

9. The expressions $y + 7$, $3y + 1$ and $4y + 3$ are consecutive terms of an arithmetic sequence. Determine y and the three terms.

10. The 25th term of the arithmetic sequence $4, 10, 16, \ldots$ is
A. 148 **B.** 150 **C.** 154 **D.** 144

11. Numerical Response The number of three-digit whole numbers that are multiples of 11 is _____.

C. Stretch

12. The expressions x^2, $4x$ and $3x + 4$ are consecutive terms of an arithmetic sequence. Determine the possible values of x and the three terms for each.

13. Sequence A is $5, 12, 19, \ldots$ and sequence B is $3, 8, 13, \ldots$
(a) Find the first two numbers that appear in both sequences.

(b) The common numbers form an arithmetic sequence. Explain why its common difference is 35.

(c) How many numbers less than 1000 appear in both sequences?

Extra Practice 3: Arithmetic Sequences in Context

A. Skills

1. For each arithmetic sequence, determine d, t_1 and the general term.
(a) $t_3 = 10,\ t_{10} = 38$ (b) $t_5 = -4,\ t_{13} = -44$ (c) $t_{10} = 6.5,\ t_{30} = 16.5$

2. An arithmetic sequence has $t_7 = 40$ and $t_{12} = 65$. Determine t_8, t_9, t_{10}, t_{11} and t_1.

3. The function $f(x) = 4x - 7$ has its domain restricted to the natural numbers. Write the first four values of the range and state the common difference.

4. The points (n, t_n) of an arithmetic sequence lie on the line $t = 40 - 3n$. State t_1, d and t_{12}.

B. Applying

5. \$5000 is invested at 2.8% simple interest per year.
(a) What is the investment worth after 12 years?

(b) After how many full years is it first worth more than \$7000?

6. In a concert hall, each row has the same number of extra seats as the row in front of it. Row 3 has 30 seats and row 12 has 57 seats. The last row has 99 seats. How many rows are there?

7. Computer equipment loses the same dollar value every year. It is worth \$14 400 two years after it was bought and \$6600 eight years after it was bought.
(a) Determine the annual depreciation and the original price.

(b) What is it worth ten years after it was bought?

8. The water pressure on a diver is 1.0 atm at the surface and rises 0.1 atm for every metre of depth. A gauge records the pressure every 5 m on the way down, starting at the surface: $t_1 = 1.0$.
(a) Write the first three readings and the general term.

(b) At what depth is the reading 4.5 atm?

9. A taxi charges \$4.25 plus \$1.90 per kilometre. Which general term gives the fare, in dollars, for a trip of n km?

A. $t_n = 1.90n + 4.25$

B. $t_n = 4.25 + 1.90(n - 1)$

C. $t_n = 6.15n$

D. $t_n = 4.25n + 1.90$

10. Numerical Response An arithmetic sequence has $t_6 = 47$ and $t_{15} = 110$. The value of t_{25} is _____.

C. Stretch

11. In an arithmetic sequence, $t_3 = 2k + 1$ and $t_7 = 6k + 13$. Determine d in terms of k, and show that t_1 does not depend on k.

12. In an arithmetic sequence, $t_4 + t_9 = 50$ and $t_6 = 21$. Determine t_1 and d.

13. Phone plan A costs \$60 to start plus \$12 per month. Plan B has no start-up cost and costs \$17 per month. Write the total cost of each plan after n months, and find the first month at which plan A has cost less in total.

Extra Practice 4: Arithmetic Series

A. Skills

1. Determine the sum of each arithmetic series.
(a) $10 + 13 + 16 + \ldots$ (first 18 terms)

(b) $40 + 34 + 28 + \ldots$ (first 15 terms)

(c) $-2.5 + (-1) + 0.5 + \ldots$ (first 12 terms)

(d) $\frac{3}{4} + \frac{1}{2} + \frac{1}{4} + \ldots$ (first 9 terms)

2. Determine each sum from the first and last terms. (a) $t_1 = 5$, $t_{26} = 130$ (b) $t_1 = 48$, $t_{17} = -16$

3. Determine each sum.
(a) $7 + 12 + 17 + \cdots + 122$

(b) $-3 + (-10) + (-17) + \cdots + (-87)$

4. For the sequence $11, 8, 5, 2, \ldots$, write S_1, S_2, S_3 and S_4.

B. Applying

5. How many terms of the series $9 + 11 + 13 + \ldots$ must be added to give 240?

6. An arithmetic series has $t_1 = 1$ and $S_{12} = 276$. Determine d and t_{12}.

7. Logs are stacked in layers. The bottom layer has 24 logs and each layer has one log fewer than the layer below it. The top layer has 8 logs. How many layers are there, and how many logs?

8. An engineer's starting salary is \$42 500, with a raise of \$1800 every year.
(a) What is the salary in year 10?

(b) What are the total earnings over the first 10 years?

9. An arithmetic series has $S_n = 5n^2 - n$. Determine t_1, d and t_{12}.

10. In an arithmetic sequence $t_3 = 11$ and $t_9 = 35$. The sum of the first 20 terms is

A. 790 **B.** 820 **C.** 980 **D.** 1640

11. Numerical Response The sum of all the multiples of 7 from 7 to 196 is _____.

C. Stretch

12. The sum of the first n natural numbers, $1 + 2 + 3 + \cdots + n$, is 210. Determine n.

13. Three consecutive terms of an arithmetic sequence add to 27 and multiply to 585. Determine the terms. (Hint: call them $m - d$, m, $m + d$.)

14. An arithmetic series has $S_5 = 40$ and $S_{10} = 155$. Determine t_1 and d.

Extra Practice 5: Geometric Sequences

A. Skills

1. State the common ratio and write the next two terms.
(a) $9, 36, 144, \ldots$ (b) $250, 50, 10, \ldots$ (c) $-4, 12, -36, \ldots$

(d) $\frac{3}{8}, \frac{3}{4}, \frac{3}{2}, \ldots$ (e) $6, 6\sqrt{2}, 12, \ldots$ (f) $1.2, -0.6, 0.3, \ldots$

2. Write the general term, then determine the indicated term.
(a) $3, 21, 147, \ldots$; t_7 (b) $896, 448, 224, \ldots$; t_{10} (c) $2, -10, 50, \ldots$; t_8

3. Determine the number of terms. (a) $7, 14, 28, \ldots, 3584$ (b) $6561, 2187, 729, \ldots, 1$

4. For the sequence $3, 6, 12, \ldots$, decide whether each number is a term. If it is, state its position. (a) 3072 (b) 3000

B. Applying

5. In a geometric sequence $t_2 = 15$ and $t_5 = 405$. Determine r, t_1 and the general term.

6. In a geometric sequence $t_3 = 48$ and $t_7 = 3$. Determine the possible values of r and t_8.

7. (a) Insert two geometric means between 3 and 375.

(b) Insert five geometric means between 1 and 64.

8. The expressions $x - 3$, $x + 1$ and $x + 9$ are consecutive terms of a geometric sequence. Determine x and the terms.

9. In a geometric sequence $t_1 = 2$ and $t_3 = 50$. A possible value of t_2 is

A. 26 **B.** -10 **C.** 25 **D.** 100

10. Numerical Response The number of terms in the sequence $\frac{1}{16}, \frac{1}{8}, \frac{1}{4}, \ldots, 512$ is _____.

C. Stretch

11. The expressions $x + 2$, $3x$ and $8x$ are consecutive terms of a geometric sequence. Determine x, the terms and r.

12. A sequence has general term $t_n = 3(2)^n$. Show that it is geometric, and rewrite the general term in the form $t_n = t_1 r^{n-1}$.

13. A sheet of paper is 0.1 mm thick. Each fold in half doubles the thickness. After how many folds would it be thicker than 1 m (1000 mm)? Use a table of values.

Extra Practice 6: Geometric Growth and Decay

A. Skills

1. State the growth factor in each situation.

(a) Ticket prices rise 4% per year.

(b) A forest shrinks by 9.5% per year.

(c) A rumour's reach quadruples every day.

(d) A phone loses $\frac{3}{10}$ of its value each year.

(e) A ball rebounds to 45% of its height.

(f) A town grows by one-quarter each decade.

2. Each growth factor describes a change per year. State the percent increase or decrease.

(a) 1.035 (b) 0.88 (c) 2.5 (d) 0.995

3. Write the first four terms of each sequence, where t_1 is the starting amount.

(a) Start at 2000 and grow by 5% each period.

(b) Start at 640 and decrease by 25% each period.

4. Determine each value.

(a) \$12 000 grows by 6% per year. Its value after 5 years, to the nearest cent.

(b) 850 animals decline by 11% per year. The number after 7 years.

B. Applying

5. A library lent 42 000 items in 2025. The number of loans is falling by 3% per year. Estimate the number of loans in 2033.

6. A ball is dropped from 9 m and each bounce reaches $\frac{2}{3}$ of the height of the bounce before.

(a) Determine the height of the 4th bounce, to the nearest centimetre.

(b) If $t_1 = 6$, which term is the height of the 4th bounce? _____

7. \$8000 is invested at 2.9% per year, compounded annually. Determine its value after 15 years.

8. Caffeine has a half-life of about 5 hours in the body. A person takes in 200 mg at 7:00 a.m.

(a) How much is left at 10:00 p.m.?

(b) At what time is there first less than 10 mg left (checking at the end of each half-life)?

9. A town of 5400 people is shrinking by 2.5% per year. After how many years does the population first fall below 4000?

10. A truck worth \$48 000 depreciates by 15% per year. Which expression gives its value after 6 years?

A. $48\,000(0.15)^6$ **B.** $48\,000(0.85)^5$ **C.** $48\,000(1.15)^6$ **D.** $48\,000(0.85)^6$

11. Numerical Response \$5000 is invested at 4% per year, compounded annually. Its value after 10 years, to the nearest dollar, is _____.

C. Stretch

12. Option A: \$5000 at 4.8% per year compounded monthly. Option B: \$5000 at 4.9% per year compounded annually. Which is worth more after 3 years, and by how much?

13. A coin bought for \$250 is worth \$410 six years later. Assuming a constant yearly rate of growth, determine the rate to the nearest tenth of a percent.

14. Town A has 20 000 people and is shrinking by 3% per year. Town B has 12 000 people and is growing by 4% per year. After how many years does Town B first have more people than Town A?

Extra Practice 7: Geometric Series

A. Skills

1. Determine each sum. Give exact answers.
(a) $4 + 12 + 36 + \cdots \quad (S_9)$

(b) $1000 + 500 + 250 + \cdots \quad (S_8)$

(c) $2 - 8 + 32 - \cdots \quad (S_6)$

(d) $0.5 + 1 + 2 + \cdots \quad (S_{10})$

2. Determine each sum.
(a) $6 + 12 + 24 + \cdots + 3072$

(b) $4000 + 400 + 40 + \cdots + 0.04$

3. A geometric sequence has $t_1 = 7$ and $r = 2$. Determine t_8 and S_8.

B. Applying

4. How many terms of $3 + 15 + 75 + \cdots$ add to 11 718?

5. In a geometric sequence, $t_2 = 12$ and $t_5 = 324$. Determine S_7.

6. A geometric series has $t_1 = 3$ and $S_3 = 93$. Determine all possible values of r.

7. For a series, $S_n = 6(2^n - 1)$. Determine the first three terms and t_{10}.

8. A fundraiser raises \$25 on day 1. Each day after that it raises 20% more than the day before.
(a) How much is raised in total in the first 10 days, to the nearest cent?

(b) On which day does the total first pass \$1500?

9. The sum $1 + 2 + 4 + \cdots + 512$ is

A. 511 **B.** 1023 **C.** 1024 **D.** 2047

10. Numerical Response A ball is dropped from 10 m. Each bounce reaches 60% of the height before. The total vertical distance it has travelled when it hits the ground for the 4th time, to the nearest tenth of a metre, is _____.

C. Stretch

11. A geometric series has $r = -1$. Describe the values of S_n, and show that the formula $S_n = \dfrac{t_1(r^n - 1)}{r - 1}$ agrees.

12. A geometric series has $t_1 = 2$ and $S_4 = 5S_2$. Determine all possible values of r.

13. Ava deposits \$1000 at the start of each year into an account paying 5% per year, compounded annually. Write the value at the end of 10 years as a geometric series and find its sum, to the nearest cent.

14. Explain why the geometric series formula cannot be used directly on $10 + 3 + 2 + 4 + 8 + \cdots + 512$. Then determine the sum.

15. A sequence has $t_1 = 2$ and $t_3 = 32$, and all its terms are positive. Find the sum of the first four terms if the sequence is (a) arithmetic, (b) geometric.

Extra Practice 8: Infinite Geometric Series

A. Skills

1. Determine r and state whether each series is convergent or divergent. Calculate S_∞ where it exists.
(a) $120 + 60 + 30 + \ldots$ (b) $9 - 6 + 4 - \ldots$ (c) $\frac{1}{8} + \frac{1}{4} + \frac{1}{2} + \ldots$

(d) $1000 + 100 + 10 + \ldots$ (e) $-5 + 5 - 5 + \ldots$ (f) $0.6 + 0.48 + 0.384 + \ldots$

2. Write each repeating decimal as a fraction in lowest terms, using an infinite geometric series.
(a) $0.\overline{2}$ (b) $0.\overline{81}$ (c) $3.\overline{6}$ (d) $0.8\overline{3}$

3. Each part describes an infinite geometric series. Determine the missing value.
(a) $t_1 = 15$, $r = -0.25$; find S_∞. (b) $S_\infty = 40$, $r = 0.6$; find t_1. (c) $t_1 = 27$, $S_\infty = 18$; find r.

4. Consider the series $50 + 25 + 12.5 + \ldots$
(a) Write the first six partial sums.

(b) What value do the partial sums approach? Confirm with the formula.

B. Applying

5. A ball is dropped from a height of 12 m. Each bounce reaches $\frac{2}{3}$ of the height it fell from. If it keeps bouncing forever, what total vertical distance does it travel?

6. In an infinite geometric series, $t_3 = 18$ and $t_5 = 2$. Determine S_∞ for each of the two possible series.

7. A yearly fundraiser raised \$25 000 in its first year. Each year after that it raised 90% of the year before.
(a) How much did it raise in year 8?

(b) How much did it raise in total in the first 8 years?

(c) What is the greatest total it could ever raise if the pattern continued?

8. Consider the infinite geometric series $4 + 4(x+1) + 4(x+1)^2 + \dots$
(a) For what values of x does the series converge?

(b) Determine S_∞ when $x = -0.5$.

9. The repeating decimal $0.\overline{18}$ written as a fraction in lowest terms is
A. $\frac{2}{11}$ **B.** $\frac{9}{50}$ **C.** $\frac{1}{6}$ **D.** $\frac{2}{111}$

10. Numerical Response An infinite geometric series has first term 21 and sum to infinity 30. The common ratio, to the nearest tenth, is _____.

C. Stretch

11. The sum to infinity of a geometric series is 3 times its first term.
(a) Determine r.

(b) If $t_2 = 8$, determine S_∞.

12. An infinite geometric series has $S_\infty = 9$, and the sum of its first two terms is 5. Determine the two possible values of r and write the first three terms of each series.

13. In an infinite geometric series, the odd-numbered terms $(t_1 + t_3 + t_5 + \dots)$ add to 36 and the even-numbered terms $(t_2 + t_4 + t_6 + \dots)$ add to 12. Determine r, t_1 and S_∞.

Extra Practice 9: Unit 1 Practice Test

A second practice test for the unit. Calculators are allowed. Round as each question asks.

A. Multiple Choice

1. Which of the following is a finite arithmetic series?

A. $2 + 5 + 8 + \cdots + 59$

B. $2, 5, 8, \ldots, 59$

C. $2 + 5 + 8 + 11 + \ldots$

D. $2 + 6 + 18 + 54$

2. Which statement about the sequence $6, 6, 6, 6, \ldots$ is true?

A. It is arithmetic only.

B. It is geometric only.

C. It is both arithmetic and geometric.

D. It is neither arithmetic nor geometric.

3. The points $(2, 11)$ and $(7, -4)$ lie on the graph of an arithmetic sequence. The common difference is

A. -15 **B.** -3 **C.** $-\frac{1}{3}$ **D.** 3

4. In an arithmetic sequence, $t_4 = 2$ and $t_9 = 22$. The value of t_{20} is

A. 62 **B.** 66 **C.** 70 **D.** 86

5. In a geometric sequence, $t_1 = 81$ and $t_5 = 1$. The possible values of r are

A. $\frac{1}{3}$ only **B.** $\pm\frac{1}{3}$ **C.** ± 3 **D.** $-\frac{1}{3}$ only

6. A laptop bought for \$1800 loses 25% of its value each year. After how many full years is it first worth less than \$300?

A. 4 **B.** 6 **C.** 7 **D.** 8

7. A town of 15 000 people grows by 2% per year. Its population P after n years can be modelled by

A. $P = 15\,000(0.02)^n$

B. $P = 15\,000(1.02)^{n-1}$

C. $P = 15\,000(1.02)^n$

D. $P = 15\,000(1.2)^n$

8. The sum of the first 6 terms of $2 - 6 + 18 - \ldots$ is

A. -486 **B.** -364 **C.** 364 **D.** 1094

9. Which infinite geometric series has a sum of 10?

A. $5 + 2.5 + 1.25 + \ldots$

B. $20 - 10 + 5 - \ldots$

C. $8 + 2 + 0.5 + \ldots$

D. $2 + 4 + 8 + \ldots$

10. A circle has a radius of 9 cm. A second circle has a radius $\frac{2}{3}$ as long, a third has a radius $\frac{2}{3}$ of the second's, and so on forever. The sum of the circumferences of all the circles is

A. 18π cm **B.** 27π cm **C.** 54π cm **D.** 81π cm

B. Numerical Response

11. Numerical Response In the arithmetic sequence $48, 45.5, 43, \ldots$, the first negative term is t_n. The value of n is _____.

12. Numerical Response In the geometric sequence $0.75, 1.5, 3, \ldots$, the number 1536 is the term t_n. The value of n is _____.

13. Numerical Response A student saves \$15 in the first week, and each week saves \$3 more than the week before. The total saved, in dollars, after 26 weeks is _____.

14. Numerical Response The sum of the first 10 terms of $0.2 + 0.6 + 1.8 + \ldots$, to the nearest whole number, is _____.

15. Numerical Response An infinite geometric series has $t_1 = 18$ and $S_\infty = 24$. The common ratio, to the nearest hundredth, is _____.

C. Written Response

16. An auditorium has 16 seats in the first row, and each row has 2 more seats than the row in front of it. There are 1066 seats in total.

(a) How many rows are there?

(b) How many seats are in the last row?

17. A patient is given 120 mg of a medicine. Each hour, the body removes 35% of the medicine present at the start of that hour.

(a) Determine the amount left after 1, 2 and 3 hours (to the nearest hundredth of a milligram).

(b) Let t_n be the amount left after n hours. Write the general term.

(c) How much is left after 5 hours?

(d) After how many full hours is the amount first less than 5 mg?

18. On December 1 the ice on a lake is 8 cm thick. In the first week the ice gets 6 cm thicker, and each week after that it gets thicker by 80% of the previous week's increase.

(a) By how much, to the nearest hundredth of a centimetre, does the ice thicken in week 4?

(b) How thick, to the nearest hundredth of a centimetre, is the ice after 6 weeks?

(c) If the pattern continued, what thickness would the ice approach?

19. The expressions $x + 1$, $x + 5$ and $x + 17$ are consecutive terms of a geometric sequence.

(a) Determine x and the three terms.

(b) If these are the first three terms, determine the sum of the first 8 terms of the series.